Optimizing Composite 3D Printed Running Shoe Plates as a Replacement to Expensive Carbon Fiber Standards

Introduction

Current running shoes utilize a plate inside the sole to increase energy return in each step during high intensity workouts and races. The primary material utilized for plating is carbon fiber; however, this comes with high manufacturing costs increasing the price of these shoes. An average carbon plated shoe costs around $200-$250 with the higher quality options costing upwards of $300 (Men's Carbon Plated Shoes, n.d.).

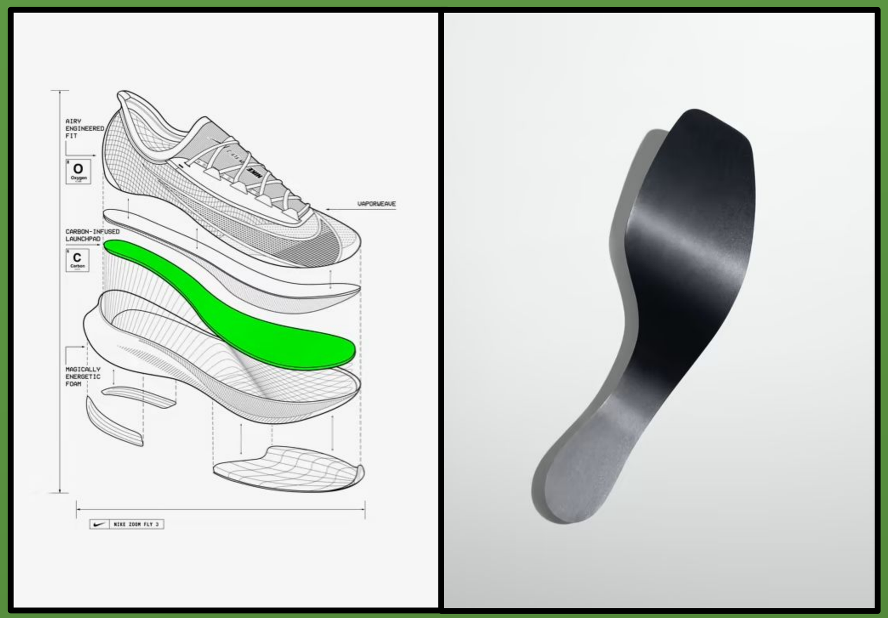

Figure 1 & 2 Nike's plate design and implementation into the Nike Zoom Fly (*Nike Zoom Fly 3*, n.d.).

While alternative materials have been tested and implemented into various running shoes, these materials, such as nylon and TPU, have primarily been used for their support as opposed to being an aggressive plate like carbon fiber. The goal of these plates is to improve a runner's running economy, the measure of how efficiently a runner uses their energy. The primary test of running economy is a VO_2 assessment which is a laboratory test that measures the amount of oxygen that is taken in while running at a set submaximal pace.

Purpose

3D printed carbon fiber composites were tested as an affordable alternative to the current carbon fiber plates. The materials tested were TPU + carbon fiber, Nylon (PA6) + carbon fiber, and Polycarbonate (PC) + carbon fiber. The plates designed are based off of a Brooks Ghost trainer and optimal shapes that have already been thoroughly researched and perfected.

If the materials are proven to be effective and improve running economy they could be implemented into a more beginner friendly shoe for high intensity training. This will allow for a lower initial cost for new runners to have the equipment necessary for them to perform at the same levels as other runners while saving manufacturing costs.

Methods

Figure 3 & 4 Brooks Ghost cut open. (Image taken by finalist, 2025)

- A Brooks Ghost was chosen because it is a neutral trainer used by many runners.
- Midsole dimensions were gathered along with optimized thickness and radius of curvature.
- Digital designs created using CAD software.

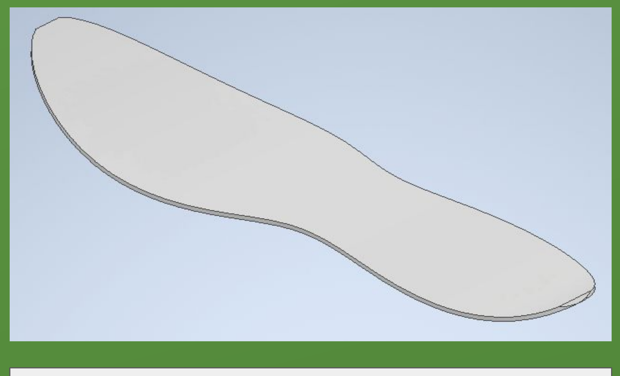

Figure 5 Final digital design. (Design created by finalist using Inventor, 2025)

Figure 6 Printing process. (Image taken by finalist, 2025)

- Different angles and print settings were tested on PLA sample plates.
- Plates were printed out of each testing material: CF+PA6, CF+PC, and CF+TPU
- CF+TPU was not a viable option due to it being significantly too flexible.

Mechanical Testing

- Plates were compressed 15mm while gathering force displacement data.
- The plates were put in between the insole and the outsole of the shoe for a second round of tests.
- The shoe was tested without plates as a control.

Figure 7 Mechanical testing process (just plate). (Image taken by finalist, 2026)

Figure 8 VO_2 test. (Image taken by finalist, 2026)

Biomechanical Testing

- VO_2 testing was conducted as a baseline without the plates.
- Three days later the same test was performed with the CF+PA6 plates in between the insole and outsole of the shoes.
- Oxygen intake data was gathered for both tests.

Results

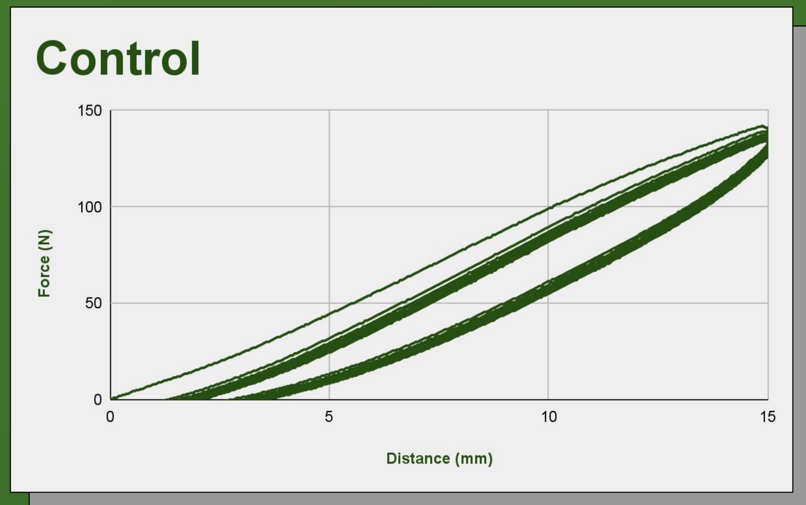

Figure 9 Force displacement graph of shoe without plate. (Graph created by finalist in Sheets, 2026)

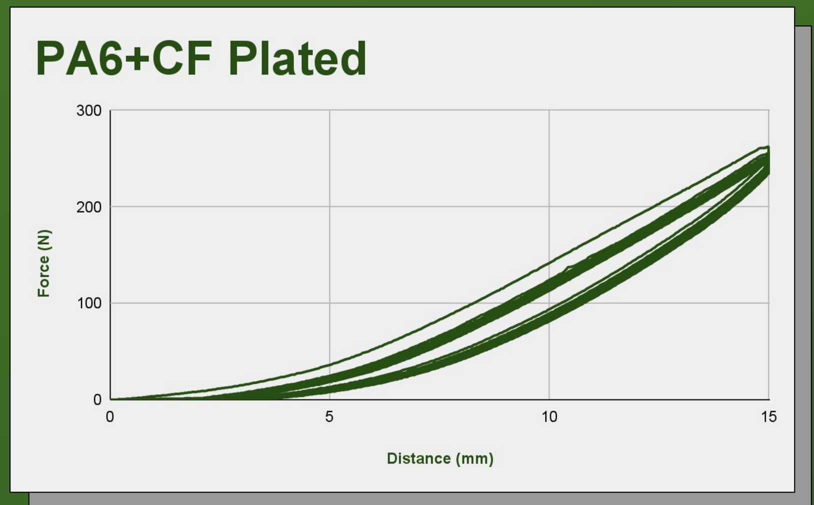

Figure 10 Force displacement graph of shoes with plates. (Graph created by finalist in Sheets, 2026)

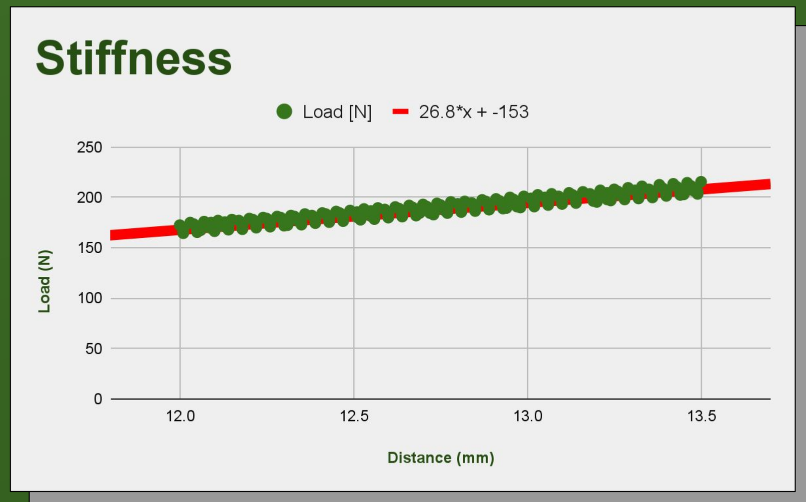

Figure 11 80-90% of the loading displacement (plated shoes). (Graph created by finalist in Sheets, 2026)

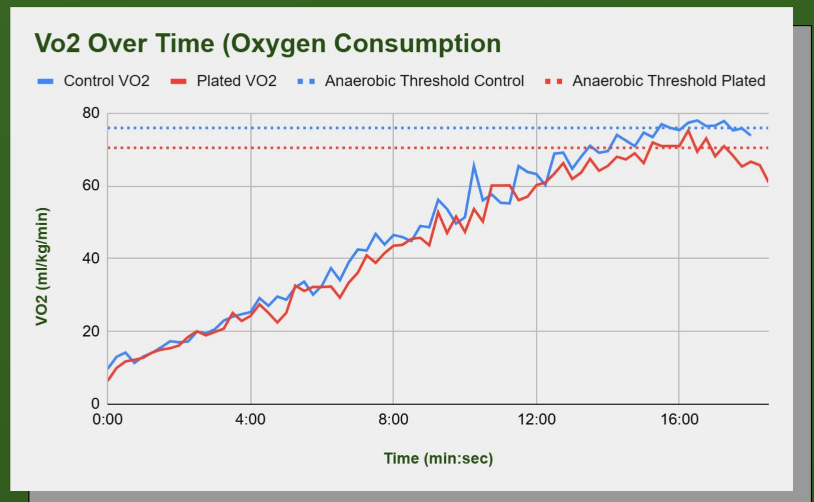

Figure 12 Comparison of both VO_2 tests. (Graph created by finalist in Sheets, 2026)

Conclusion

Mechanical and biomechanical tests proved the plates were a success. The three point bending test narrowed down the optimal material to be PA6+CF. They had a higher energy return when the plates were inserted, and the stiffness value, 26.8 N/mm, can be compared to that of an actual carbon plated shoe, 53.8 N/mm. Figure 12 shows that the oxygen intake with the plates inserted was consistently below the intake values without the plates. The VO_2 at anaerobic threshold can also be compared, 75.93 ml/kg/min without plates and 70.46 ml/kg/min with plates. These data points prove that running economy was improved by the 3D printed plates and significantly lower the manufacturing costs.

Extension

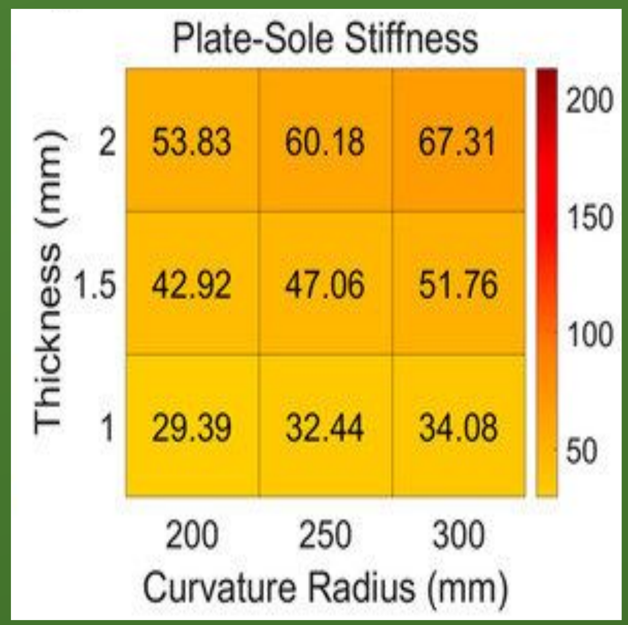

Figure 13 Plate-sole stiffness data used to compare my results, further testing would allow for more comparisons (Guo et al., 2025).

Further testing will be conducted to continue verifying the plates are effective. If the plates are intended to continue being used as an insert, stability needs to be improved due to their flat nature. Another option is to imbed the plates into the outsole of the shoe directly under the midsole for better performance. This would allow for a more accurate comparison to actual carbon fiber plated shoes. As well, more radii and thicknesses would be tested as more comparison points for previously tested carbon fiber plates.

Citations

Guo, Y., Jia, Y., Wu, Y., & Zhu, X. (2025, March 5). *Longitudinal bending stiffness analysis of composite carbon plates and shoe sole, based on three-point bending test*. MDPI. https://www.mdpi.com/2076-3417/15/5/2785

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